

Short Essay Questions – Medical Sonography

1. *What is ultrasound?*
2. *Why is ultrasound suitable for medical diagnosis?*
3. *What is medical sonography?*
4. *What is a sonogram?*
5. *What are the main steps in ultrasound image formation?*
6. *What role does the transducer play in ultrasound imaging?*
7. *Why is ultrasound gel used during scanning?*
8. *How does tissue density affect ultrasound transmission?*
9. *What is compressibility in tissues?*
10. *Why does ultrasound travel faster in bone than in soft tissue?*
11. *What is ultrasound beam intensity?*
12. *What is attenuation in ultrasound imaging?*
13. *Why does deeper tissue appear darker on ultrasound images?*
14. *What medical areas commonly use sonography?*
15. *How is ultrasound used in obstetrics?*
16. *What role does ultrasound play in gynecology?*
17. *How does ultrasound help in cardiovascular diagnosis?*
18. *What are the main advantages of ultrasound over X-rays?*
19. *Why is ultrasound considered safe for pregnant women?*
20. *What is a piezoelectric crystal?*
21. *Why are piezoelectric crystals important in ultrasound probes?*
22. *What is the function of the probe housing?*
23. *What determines the frequency of ultrasound produced by a crystal?*
24. *What is acoustic impedance?*

25. *Why does reflection occur at tissue boundaries?*
26. *What happens when ultrasound strikes air?*
27. *What is refraction in ultrasound imaging?*
28. *How does beam angle affect image quality?*
29. *What is an ultrasound echo?*
30. *How is echo information used to create an image?*
31. *What is image display in ultrasound?*
32. *What does grayscale represent in ultrasound images?*
33. *What is absorption in ultrasound physics?*
34. *Why is absorption higher in bone than in soft tissue?*
35. *How does tissue viscosity affect ultrasound transmission?*
36. *Why is ultrasound frequency selection important?*
37. *What frequencies are typically used for deep abdominal imaging?*
38. *What frequencies are used for superficial structures?*
39. *What is Doppler sonography?*
40. *What physical principle is Doppler ultrasound based on?*
41. *Why is Doppler ultrasound important in medicine?*
42. *What is color Doppler imaging?*
43. *What is power Doppler imaging?*
44. *What is spectral Doppler imaging?*
45. *What is duplex Doppler imaging?*
46. *How is Doppler ultrasound used in obstetrics?*
47. *How does Doppler help detect vascular disease?*
48. *What is continuous-wave Doppler?*
49. *What is pulsed-wave Doppler?*
50. *Why is pulsed Doppler preferred for localized measurements?*

51. *What is aliasing in Doppler ultrasound?*
52. *How does ultrasound assist in detecting tumors?*
53. *What role does ultrasound play in eye examinations?*
54. *How does ultrasound help evaluate the liver?*
55. *Why is ultrasound useful for gallbladder imaging?*
56. *What is musculoskeletal ultrasound used for?*
57. *What advantages does portable ultrasound provide?*
58. *Why is real-time imaging important in ultrasound?*
59. *What is echocardiography?*
60. *How does ultrasound help in detecting fluid collections?*
61. *Why is ultrasound useful for guided procedures?*
62. *What factors affect ultrasound image quality?*
63. *Why does bone appear bright on ultrasound images?*
64. *Why do fluids appear dark on ultrasound images?*
65. *What is the importance of transducer design?*
66. *How does ultrasound contribute to emergency medicine?*
67. *Why is ultrasound preferred for pediatric imaging?*
68. *What is the role of ultrasound in cancer detection?*
69. *How does ultrasound support prenatal care?*
70. *What makes ultrasound a versatile diagnostic tool?*